

A study of object detection in histopathology

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Abstract: Medical image analysis has become increasingly important for automated medical diagnosis, as well as deep learning. Specifically, object detection models may help in automatically identifying pathological structures and features. This study presents a comprehensive comparative analysis for object detection tasks in histological images of the latest models including the YOLO (You Only Look Once) architectures, from YOLOv8 to the recently introduced YOLOv12. These models were evaluated alongside alternative architectures including RT-DETR, YOLO-World, and YOLOE across five diverse histology datasets: BCNB, Nuclei, TNBC, MoNuSAC, and CryoNuSeg. The experimental analysis employed standardized training protocols with consistent hyperparameters and data augmentation strategies, evaluating the performance through multiple metrics, inference time, and computational cost. The results obtained on the five datasets indicate that YOLOv11 consistently showed a strong performance across multiple datasets, however the newly introduced attention mechanisms of YOLOv12 show good performance, despite the model having slightly lower overall performance. Specialized variants like YOLOE demonstrated promising results for specific applications, while RT-DETR showed poor performance on smaller objects, which are typical in histological images. This comprehensive evaluation provides insights for the current state of object detection architectures for clinical histopathology applications and establishes benchmarks for future avenues of research in automated medical image analysis.

Keywords: YOLO, object detection, histology, deep learning, medical image analysis, pathology, computer vision

1 Introduction

1.1 Histology

Human pathologists perform diagnoses by analysing, through a microscope, a treated and stained specimen [1] on a glass slide [2]. This process is called histopathology [1] while the images used in the process are called histopathological images [1]. Through specialised hardware, the entire slide may be captured with a scanner and saved as a high-resolution digital image, also called a Whole Slide Image (WSI) [2]. The digitization of histopathology has revolutionized the field, enabling the development of computational approaches for automated analysis and diagnosis. To visualize different components of the tissue under a microscope, the sections are first treated and then dyed with one or more stains, with Hematoxylin and Eosin (H&E) being the most commonly used staining technique [1]. The histological images are used for the accurate diagnosis of different diseases, such as cancers, enabling the visualization of the complexity at multiple levels, enabling the analysis of intricate tissue structures, diverse cellular morphologies, and pathological changes that guide treatment decisions.

Manual annotation of histological structures is time consuming and subject to inter-variability of experts [3], [4], as well as heavily reliant on expert knowledge. These challenges are a motivation for the development of automated solutions that can provide consistent, rapid and objective analysis of histological specimens. These same complexities of histological images present unique challenges for computer vision algorithms.

Databases of histological images have been made publicly available which spiked the attempt to apply deep learning techniques to automate the analysis of structures present in histological slides [3], [5], [6], [7]. These deep learning approaches have shown promising results in pathology image analysis [8], [9], [10], [11] on various tasks including cell detection, nucleus detection [12], [13], [14], tissue segmentation, and cancer diagnosis [15], [16], demonstrating the potential to augment pathologist expertise, improve diagnostic accuracy and decrease the inherent time-cost.

1.2 Neural networks in medical applications

The research community has identified the potential of deep learning algorithms for biomedical image analysis due to their efficient processing of complex data [12], [17]. The advancements in both hardware and deep learning architectures present an unprecedented potential to advance medical image analysis [12], with segmentation and object detection models showing particular promise [13], [14], [16] in general and for histopathological applications [8], [9], [10], [11]. Neural networks have been increasingly used in medical applications in recent years [18], [19] with newer and more performant approaches being created [30]. This is also supported by the substitution of image recognition techniques with deep learning approaches [20], [21], [22] including in pathological image analysis. Among these deep learning approaches, segmentation and object detection models have emerged as particularly useful for biomedical images analysis [12] with a three-fold increase between 2018 and 2022 as the third most popular task in medical images [19].

Several studies have demonstrated the effectiveness of object detection and segmentation models. In nucleus detection/segmentation with fluorescence microscopy, YOLO and U-Net [13] have been used with promising results. A combination of U-Net models has been developed for automatic tissue segmentation [7]. In OCT scans, U-Net and U-Net++ [23], models were used for the identification of biomarkers through the segmentation of retinal layers and fluids.

Object detection architectures can be divided into three categories: points-based detection models, two-stage detection models and single-stage detection models [24]. Perhaps the most known models throughout the large number of architectures are the "You Only Look Once" (YOLO) [24] family of models. The YOLO models have seen a great number of variants and versions through the years with new internal mechanisms for general improvements in performance or for specific tasks. Their notoriety can be attributed to their single-stage approach to detection which provides a balance between speed and accuracy, an important requirement for real-time analysis, as is the case in the biomedical field.

The YOLO architectures have been demonstrated to have applicability in a variety of medical tasks. YOLOv4 was applied for the nucleus detection in breast histopathological images [14]. YOLOv5 has shown a high performance for coronary OCT [25]. A modified YOLOv7 outperformed other traditional methods in the automated detection of blood vessels [26]. YOLOv8 and YOLOv9 have shown progress in feature detection for diabetic retinopathy from retinal fundus images with some limitations [27].

Several literature reviews [1], [8], [28], [29] have discussed the analysis of histopathological images including the use of machine learning algorithms. Machine learning has been identified as a promising avenue of research in histopathology due to its ability to uncover subtle patterns invisible to the human eye [15], [30] with the potential to trace cancer origins in histopathological images [15], [31]. Other studies [12] have identified the YOLO family of models as the most frequently used ranging across a wide variety of medical applications such as brain tumor detection, glaucoma detection, pulmonary embolism detection, skin lesion detection, kidney disease localization and many others [12]. The YOLO family of models was also identified as the most performant due to their robustness, high accuracy, and low computational cost [12].

YOLO has also been previously suggested and applied to histology [12]. YOLOv3 achieved a high performance in the detection of prostate cancer in histological images [32]. YOLOv4 was used in nucleus detection in breast histopathological images [14], as well as YOLOv5 [16]. A modification of YOLOv5 called Histology-based Detection using Yolo (HD-Yolo) [11] was designed for nucleus segmentation and tumor microenvironment characterization. YOLOv8 was applied for cochlea detection (using bounding boxes for localization) in macaque histology [32]. A hybrid pipeline was developed for fast cell detection using YOLOv11 and StarDist [33] for cell detection and segmentation that was shown to be highly performant on both both proprietary data and the public MoNuSeg benchmark dataset [34].

A comparative analysis was designed to evaluate the more traditional object detection architectures based on CNNs against newer ones based on attention mechanisms. A selection of highly performant object detection models from the latest YOLO versions from YOLOv8-v12 against

alternative architectures such as RT-DETR, YOLO-World, YOLOE were evaluated. These models were evaluated on five diverse histological datasets from the perspective of five performance metrics (mAP@50, mAP@50-95, F1 score, precision and recall) and computational efficiency (inference time, FLOPs). A standardized parametrization was used in the analysis to guarantee a fair comparison including hyperparameter optimization and data augmentation.

Our contributions include: (1) benchmarking of the latest YOLO versions from YOLOv8-v12 against alternative architectures such as RT-DETR, YOLO-World, YOLOE across five diverse datasets for object detection of cancerous regions in histopathological images, (2) an analysis of performance trade-offs between accuracy, speed, and computational efficiency in the clinical context of the data, and (3) insights for the deployment of automated histology analysis systems in clinical settings.

2 Materials and Methods

2.1 Datasets

We selected five histology datasets to comprehensively evaluate the performance of object detection models with a focus on breast tissue images with H&E staining, yet datasets including other organs have been included as well. All annotation types have been converted to YOLO bounding boxes to ensure compatibility with the object detection paradigm and consistent evaluation across all models. The datasets represent a range of clinical scenarios and technical challenges commonly encountered in computational analysis of histopathological data. A small description relating the characteristics of each dataset is shown in Table 1.

Scalebars are not included in the histological images presented given that: 1) the images have previously been published, as indicated, and are used for feature detection and 2) we were unable to obtain information regarding the magnification of the images in two of the datasets (BCNB [35] and CryoNuSeg [38]).

Table 1 – Short description of datasets.

Dataset	Organs	Staining	Annotation type	Image count	Classes
BCNB [35]	Breast	H&E	JSON polygon vertices	1058	1
Nuclei [10]	Breast	H&E	PNG binary mask	143	1
TNBC [36]	Breast	H&E	PNG binary mask	50 (from 11 patients)	1
MoNuSAC [37]	Breast Lung	H&E	XML region vertices	209	4 names:

	Prostrate Kidney				[Epithelial, Lymphocyte, Neutrophil, Macrophage]
CryoNuSeg [38]	Adrenal gland Larynx Lymph nodes Mediastinum Pancreas Pleura Skin Testes Thymus Thyroid gland	H&E	PNG binary mask	30 (3 per organ)	1

2.1.1 BCNB

BCNB [35] dataset comprises digitized H&E-stained WSIs of primary tumor core-needle biopsy (CNB) specimens for predicting axillary lymph node (ALN) metastasis in early breast cancer (EBC). This extensive collection includes data from 1,058 EBC patients with clinically negative ALN, enrolled between May 2010 and August 2020. To ensure the dataset's quality and relevance for deep learning applications, strict inclusion criteria were applied, such as pathologically confirmed primary invasive breast cancer, comprehensive baseline clinicopathological data, and adequate biopsy material. The dataset was systematically divided, with 840 patients allocated to the training cohort and 218 patients to an independent test cohort, alongside a validation cohort comprising 25% of the training data. The JSON-based polygons have been automatically converted to bounding boxes for the object detection models. An example of conversion can be found in Figure 1.

Due to the large size of the WSI histological images, a compression must be made to increase the efficiency and reduce both the storage size and inference time of neural network models [39]. Downsampling of 16x has been shown to improve the precision of object detection models while slightly lowering recall on renal WSI histological images [39], with optimal downsampling being at 8x. Compression is another available option for further downsizing the high resolution images [39], with an optimal rate of 40%. Furthermore, lower resolution images have been shown to beneficially impact performance of segmentation models [40]. All images have been downsampled 20x due to their large size and computational cost.

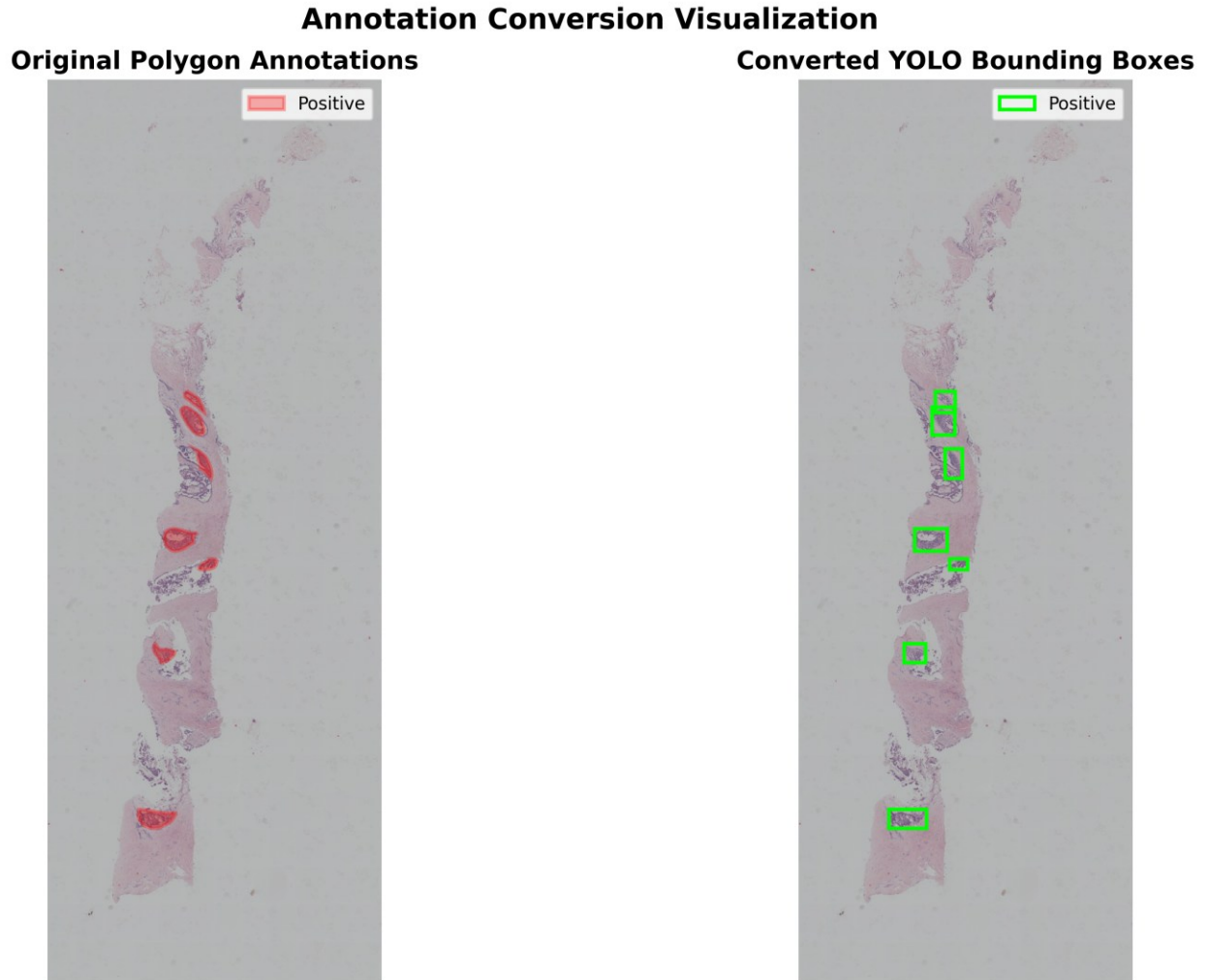


Figure 1 – Example from the BCNB [35] dataset with the original annotations (left) and the converted yolo bounding boxes (right).

2.1.2 Nuclei

Nuclei [10] dataset comprises 143 2,000 x 2,000 images ER+ BCa images scanned at 40x with 12,000 nuclei manually segmented across 137 patients. The segmentation masks have been automatically converted to bounding boxes for the object detection models. An example of conversion can be found in Figure 2.

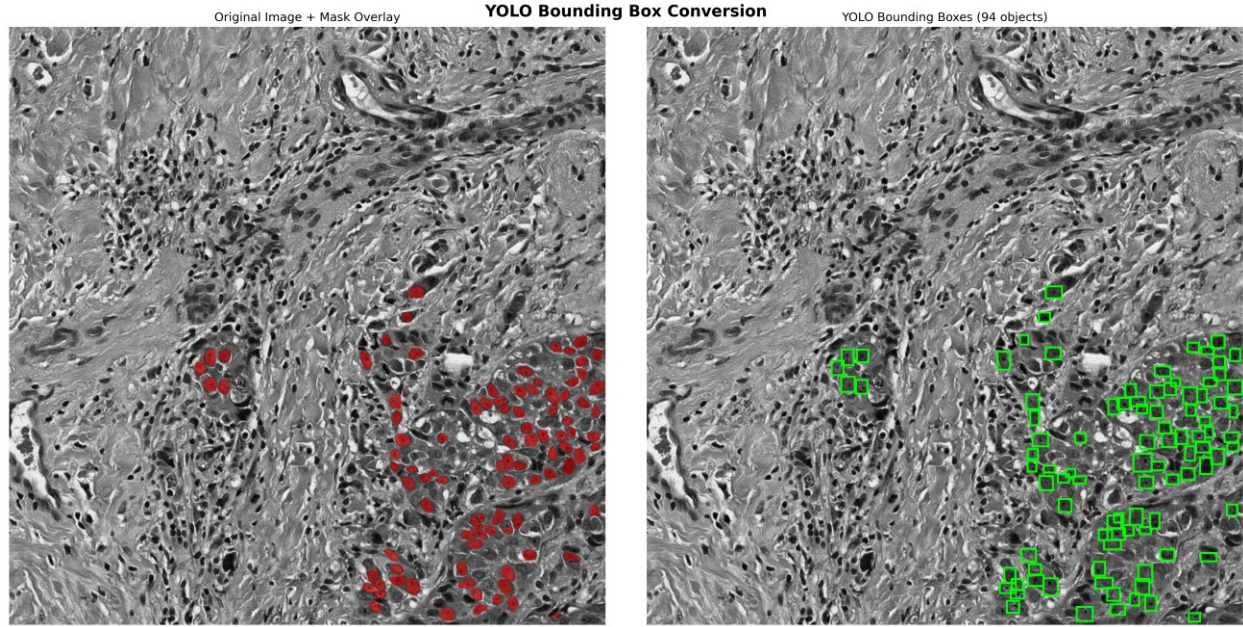


Figure 2 – Example from the Nuclei [10] dataset with the original annotations (left) and the converted yolo bounding boxes (right).

2.1.3 TNBC

TNBC [36] dataset was generated at the Curie Institute, comprising 50 images from 11 patients with a total of 4,022 annotated cells from H&E stained histological slides at 40x magnification. These slides were sourced from Triple Negative Breast Cancer (TNBC) patients. The segmentation masks have been automatically converted to bounding boxes for the object detection models. An example of conversion can be found in Figure 3.

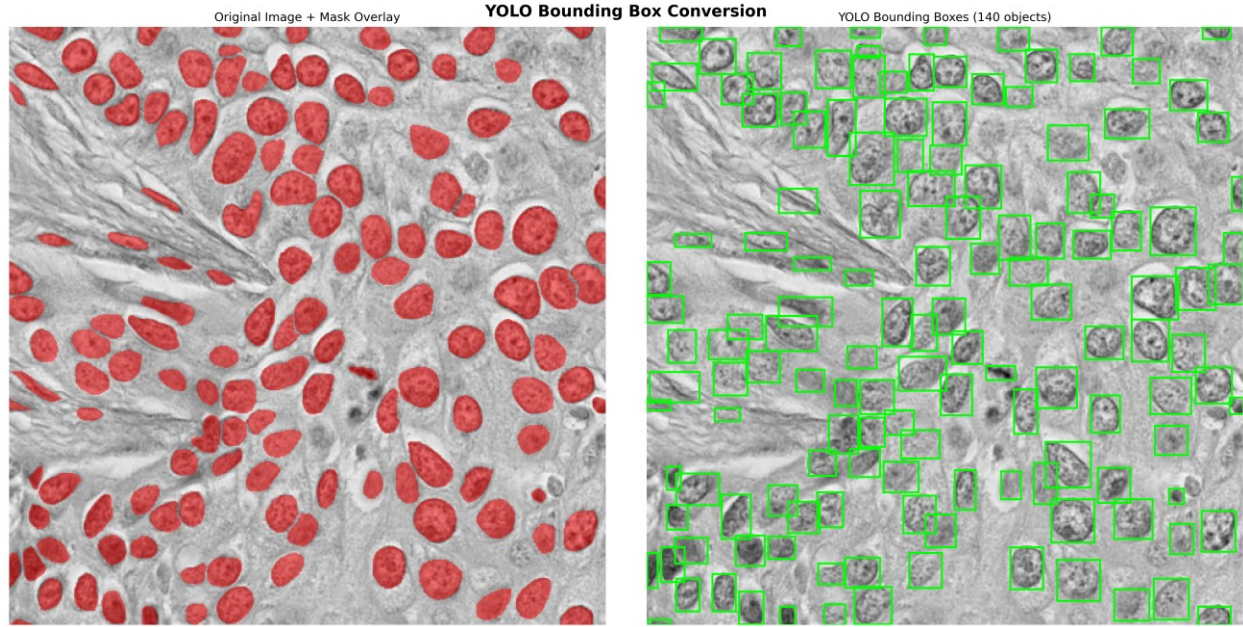


Figure 3 – Example from the TNBC [36] dataset with the original annotations (left) and the converted yolo bounding boxes (right).

2.1.4 MoNuSAC

The MoNuSAC [37] dataset is a comprehensive, multi-organ dataset specifically designed for nuclei segmentation in histopathological images to encourage the computer vision research community to develop and test algorithms for detecting, segmenting, and classifying nuclei, tasks crucial for characterizing the tumor microenvironment (TME) in cancer prognostication and research. This large and diverse dataset comprises over 46,000 hand-annotated nuclei, sourced from 71 patients across 31 hospitals. It includes nuclei from four different organs and four distinct nucleus types: epithelial cells, lymphocytes, macrophages, and neutrophils. The annotations were reviewed by an expert pathologist, with iterative revisions ensuring less than 1% error in annotations. The XML-based polygons have been automatically converted to bounding boxes for the object detection models. An example of conversion can be found in Figure 4.

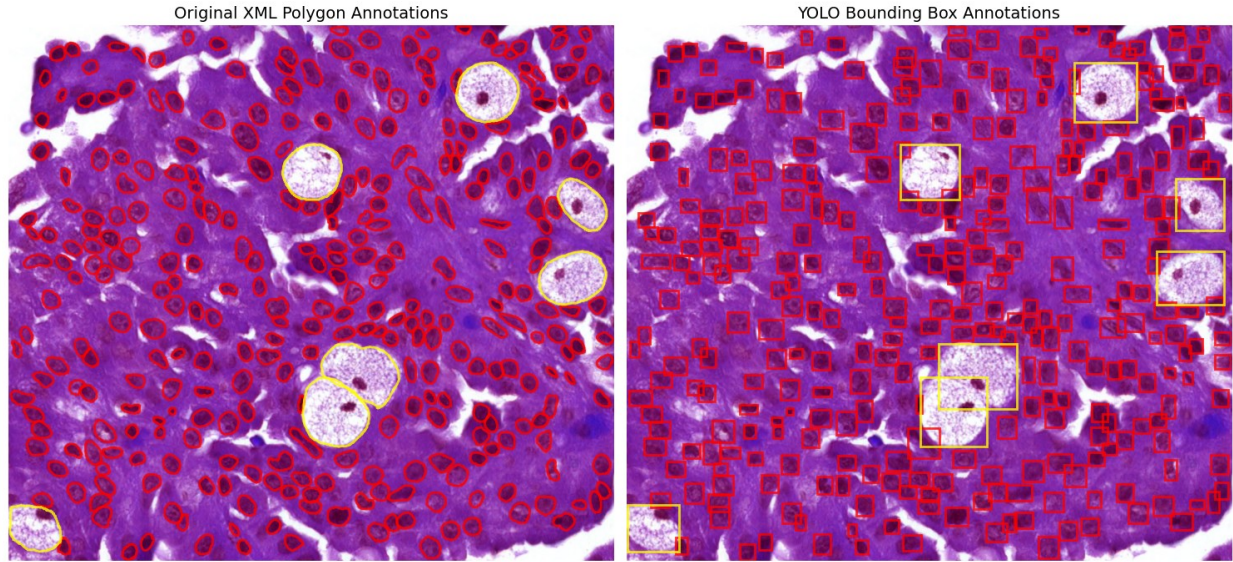


Figure 4 – Example from the MoNuSAC [37] dataset with the original annotations (left) and the converted yolo bounding boxes (right).

2.1.5 CryoNuSeg

CryoNuSeg [38] is manually annotated nuclei instance segmentation dataset derived exclusively from frozen section (FS) H&E-stained images. CryoNuSeg comprises images from 10 human organs that were not exploited in other publicly available datasets, specifically the adrenal gland, larynx, lymph node, mediastinum, pancreas, pleura, skin, testis, thymus, and thyroid gland. It includes 30 whole slide images (WSIs), from which 512x512 pixel image patches acquired at 40x magnification were extracted, with one patch per WSI. The binary segmentation masks have been automatically converted to bounding boxes for the object detection models. An example of conversion can be found in Figure 5.

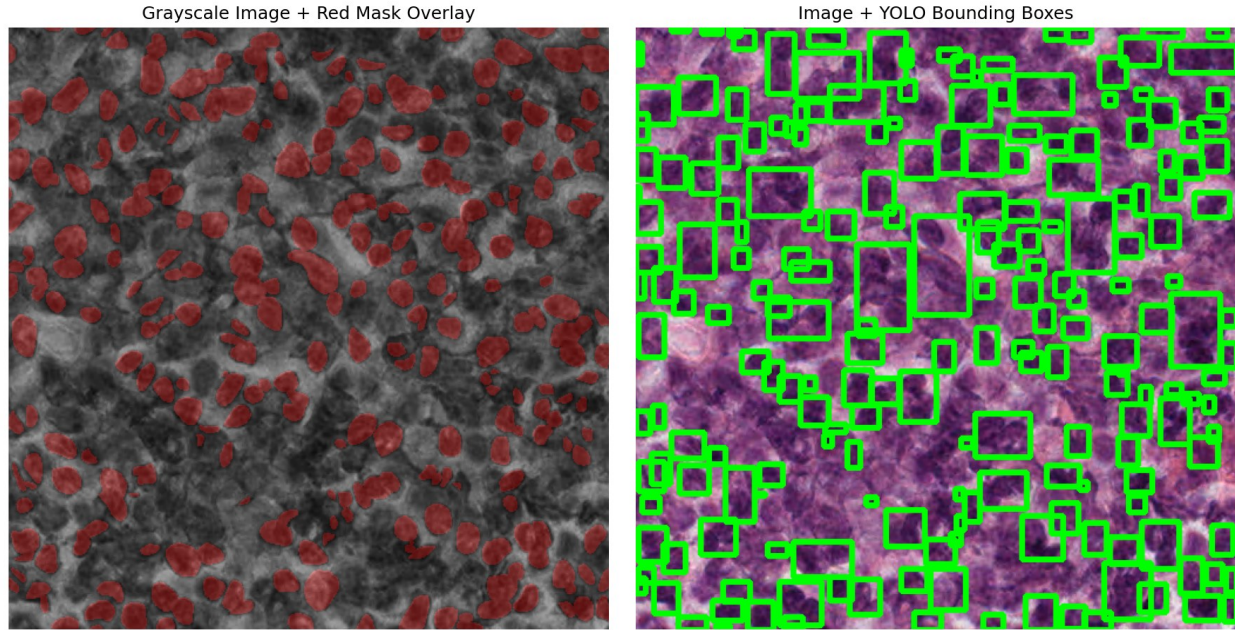


Figure 5 – Example from the CryoNuSeg [38] dataset with the original annotations (left) and the converted yolo bounding boxes (right).

2.2 Object detection models

The "You Only Look Once" (YOLO) [24] algorithm revolutionized real-time object detection through the unification of region proposal and classification. YOLO divides an image through a grid attempting to predict a class and a bounding box for each cell of the grid. This approach significantly reduced computation time and enabled end-to-end learning. YOLOv1 managed to achieve real-time detection although it incorporated a CNN backbone; however, small objects were hard to identify and localize accurately due to the fact that each grid cell was only able to predict two bounding boxes.

Since the introduction of YOLO in 2015, the YOLO family of algorithms [41] has expanded both in depth and width with several incremental improvements and various modified versions for specific applications. Each new version introduced advancements with the goal of improving computational efficiency and detection accuracy. YOLOv2 [41], [42], [43] brought several enhancements to YOLOv1 including higher input resolution, batch normalization, and the adoption of anchor boxes from Faster R-CNN to improve training and performance. YOLOv3 [41], [42], [44] achieved higher accuracy in small-object detection through multi-scale predictions using Feature Pyramid Networks (FPN) and a deeper architecture, specifically Darknet-53. YOLOv4 [41], [42], [45] managed to set new standards for detection accuracy and inference speed through several new introductions, such as: CSPDarknet-53, PANet, Mish activation, and data augmentation methods such as Mosaic and CutMix. YOLOv5 by Ultralytics [41], [42], [46] was designed with a PyTorch-based framework with an emphasis on usability and performance and made several changes such as: enhanced backbone, neck, and head designs, and an auto-anchor

learning mechanism that optimized anchor sizes during training. YOLOv6 [41], [42], [47] added the EfficientRep backbone and RepOptimizer to improve efficiency. YOLOv7 [41], [42], [48] improved efficiency by introducing a trainable bag-of-freebies and the Extended Efficient Layer Aggregation Network (E-ELAN).

YOLOv8 [41], [42], [46], also from Ultralytics [46], integrated the C2f module into the architecture and transitioned to anchor-free detection for a more simplified and specialized detection. YOLOv9 [41], [42], [49] incorporated the Generalized Efficient Layer Aggregation Network (GELAN) and Programmable Gradient Information (PGI) to mitigate information degradation in deep networks. YOLOv10 [41], [42], [50] removed non-maximum suppression (NMS) and introduced a dual assignment strategy and lightweight classification heads achieving real-time efficiency. YOLOv11 [41], [42], [46] integrated the C3k2 block, SPPF, and C2PSA into the architecture to improve feature extraction and spatial attention while balancing precision and efficiency.

YOLOv12 [41], [42], [46], [51] shifted toward an attention-centric architecture achieving state-of-the-art accuracy and efficiency while maintaining real-time performance [42], [51]. The innovations that allowed YOLOv12 to achieve this (confirmed through ablation studies [51]) are: the Area Attention module, which reduces computational complexity while maintaining a large receptive field, and Residual Efficient Layer Aggregation Networks (R-ELAN) that enhance feature aggregation and optimize training for larger models. Other architectural updates include FlashAttention for memory optimization, MLP ratio adjustments, attention-convolution integration for efficiency, removal of positional encoding, and the Position Perceiver using large separable convolutions for positional awareness. Extensive analyses confirmed YOLOv12's performance [51] showing increased accuracy, speed, and reduced computational cost compared to YOLOv6, YOLOv8, YOLOv10, YOLOv11, and RT-DETR variants. Despite its recent introduction in February 2025 [51], YOLOv12 has been successfully applied to early detection of sexually transmitted diseases [52], UAV tracking of [53], fruit detection [54], [55], and marine litter detection [56], [57] outperforming earlier YOLO versions.

YOLO-World [58] was derived from YOLOv8 and it added open-vocabulary detection using vision-language modeling demonstrating increased performance in real-world applications and in zero-shot scenarios for a wide range of objects. YOLO-World allows for the interaction between visual and linguistic information through the Re-parameterizable Vision-Language Path Aggregation Network (RepVL-PAN) and the alignment of this information through region-text contrastive loss, enhancing zero-shot detection for diverse applications.

YOLOE [59] was derived from YOLOv8 and YOLOv11 and it added prompt-based mechanisms for both object detection and instance segmentation which allows for the integration of contextual information; thus, enhancing preprocessing with context-relevant information [60]. It supports text prompts via the Re-parameterizable Region-Text Alignment (RepRTA) strategy, visual prompts via the Semantic Activated Visual Prompt Encoder (SAVPE), and no prompts via the Lazy Region-Prompt Contrast (LRPC) strategy.

Real-Time DETection TRansformer (RT-DETR) [55], [61], [62] is an alternative approach to object detection based on an end-to-end transformer removing the need for non-maximum suppression. RT-DETR replaces the standard Transformer encoder with a hybrid encoder to process intra-scale interactions and cross-scale feature fusion separately, increasing speed and detection accuracy via a query selection mechanism. RT-DETRv2 [62] achieves increased multi-scale detection through several modifications: dynamic data augmentation, scale-adaptive hyperparameters, and Selective Multi-Scale Sampling.

The YOLO family of models are released in multiple model scales offering trade-offs between accuracy, speed, computational cost, and parameter count. They are represented by suffixes: nano (n), small (s), medium (m), large (l), extra large (x). Larger models provide higher accuracy at increased computational cost, allowing users to select versions based on specific application needs, from maximum speed (n) to highest precision (x).

2.3 Performance evaluation

Precision represents the percentage of positive predictions that are correct and is calculated as the ratio of true positives (TP) to the sum of true positives and false positives (FP). Precision measures the ability to avoid false alarms and is computed as:

$$\textbf{Precision} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FP}}$$

Where *TP* represents the correct detections compared to the ground truth and *FP* represents the incorrect detections where no object actually exists.

Recall represents the percentage of actual positives correctly identified and is calculated as the ratio of true positives to the sum of true positives and false negatives (FN). Recall measures the ability to avoid missed detections and is computed as:

$$\textbf{Recall} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FN}}$$

Where *TP* represents the correct detections compared to the ground truth and *FN* represents the cases where an object is missed compared to the ground truth.

The F1 score is considered a balanced measure of performance as it computed as the harmonic mean of precision and recall. In contrast to precision and recall, the F1 score incorporates both for false positives and false negatives. This makes it particularly useful for the evaluations of performance in applications where correctly identifying positive cases is as important as minimizing false alarms. It is computed as:

$$\textbf{F1} = \frac{2 * \textbf{Precision} * \textbf{Recall}}{(\textbf{Precision} + \textbf{Recall})}$$

Intersection over Union (IoU) represents the overlap in area between predicted and true bounding boxes. IoU assesses how accurately the model is able to capture object boundaries and is computed as:

$$IoU = \frac{\textit{Intersection area}}{\textit{Union area}}$$

Mean Average Precision at an IoU threshold of 0.5 (mAP@50) represents the area under the precision-recall curve, also known as the Average Precision (AP), computed for each class where detections achieve an overlap with true bounding boxes at least 0.5 IoU. It is computed as:

$$mAP@50 = \frac{1}{|C|} \sum_{c \in C} AP_c(IoU \geq 0.5)$$

Where c represents a single class of from the set of classes C .

Mean Average Precision across IoU thresholds from 0.5 to 0.95 (mAP@50–95) represents AP scores averaged across multiple IoU thresholds ranging from 0.5 to 0.95 in increments of 0.05. This metric reflects localization performance under varying strictness levels, penalizing detections that only loosely match true objects. It is computed as:

$$mAP@50 - 95 = \frac{1}{|C|} \sum_{c \in C} \left(\frac{1}{|T|} \sum_{t \in T} AP_c(IoU \geq t) \right)$$

Where c represents a single class of from the set of classes C , and t represents a single threshold from the set of thresholds $T = \{0.5, 0.55, \dots, 0.95\}$.

Although commonly practiced, evaluating performance using a single performance metric can be risky endeavor [63], [64]. Especially if the metric fails to capture all relevant aspects, potentially leading to misleading and biased results. Therefore, we employed multiple complementary metrics to provide a more thorough evaluation across different aspects of performance.

3 Results

All models were trained using identical experimental conditions to ensure fair comparison. The training procedure used the same random subset of 70% of the original dataset for training, while validation and testing subsets contained 20% and 10%, respectively (unless specified otherwise, such as the splits made by organs or patients). All results presented are on the test set.

Models were trained for 50 epochs across the entire training set with batches of 16 images rescaled to 640×640 pixels. The AdamW optimizer was used with a starting learning rate of 1e-4, which

was reduced to $1e-6$ during the training process. A small penalty to large weights was added through a weight decay of $5e-4$. A training learning rate warmup of 10 epochs was used to stabilize training. Data augmentation was applied to reduce the possibility of overfitting. Training images in HSV color space were randomly adjusted with hue, saturation, and value shifts of 0.015, 0.7, and 0.4, respectively. Additional augmentations included random rotations of up to $\pm 10^\circ$, translations of up to 10% along either axis, and random zoom variations of up to 20%. Mixup augmentation was applied during training, blending two images into a single composite to enhance generalization by introducing additional noise and variability. Mosaic augmentation was also employed to combine multiple images, simulating visually complex scenarios. For model stability, mosaic augmentation was disabled during the final 10 training epochs.

The comparative evaluation of object detection models across five histology datasets showed considerable variations in performance depending on the dataset characteristics and models. Highlighting importance of considering multiple evaluation metrics and dataset- and domain-specific challenges when selecting object detection models for specific applications.

3.1 BCNB

The BCNB dataset [35], containing 1058 breast cancer nuclei images, presented a challenge for all evaluated models as shown in Table 2, presumably due to the small object sizes as this dataset contains the most images. YOLOv12 achieved the highest mAP@50 score of 0.285, closely followed by YOLOv10 with 0.281. However, the performance gap between different YOLO versions was relatively narrow, suggesting that the dataset characteristics may be too complex even for the advanced features of newer architectures.

The results show that RT-DETR performed poorly across all metrics, with particularly low mAP scores and extremely high inference times (1010.3 ms), making it unsuitable for real-time clinical applications. YOLOv10 demonstrated the best speed-accuracy trade-off, achieving high precision (0.519) with the fastest inference time (126.7 ms) among YOLO models despite its slightly higher computational cost of 24.5 GFLOPs, though at the cost of lower recall (0.066). YOLOv12, despite its attention mechanisms, showed only marginal improvements over YOLOv11, suggesting that this dataset may not fully benefit from the increased architectural complexity. The computational cost remained comparable between YOLOv11 and YOLOv12 (21.3 vs 21.2 GFLOPs), but YOLOv12 showed slightly higher inference time.

Table 2 – Results of object detection models on the BCNB [35] dataset.

Model/Metric	mAP@50	mAP@95	Mean F1	Mean Precision	Mean Recall	Inference Time [ms]	Computational Cost [GFLOPs]
RT-DETR	0.183	0.085	0.263	0.270	0.257	1010.3	103.4
YOLOv8	0.231	0.126	0.200	0.376	0.136	210.9	28.4
YOLOv9	0.268	0.143	0.166	0.469	0.101	205.8	26.7

YOLOv10	0.281	0.151	0.117	0.519	0.066	126.7	24.5
YOLOv11	0.266	0.147	0.247	0.411	0.176	148.2	21.3
YOLOv12	0.285	0.155	0.226	0.460	0.150	150.2	21.2
YOLOE	0.243	0.123	0.265	0.340	0.218	169.6	35.3
YOLOv8-Worldv2	0.252	0.141	0.191	0.425	0.123	135.6	32.6

The visual inspection of results shown in Figure 6 suggests that none of the models were capable of learning the correlations between the bounding boxes and the small-scale features of the images. With a subset of models introducing false positives that are not localized near the pathological features, or even not being able to predict any bounding boxes for certain images.

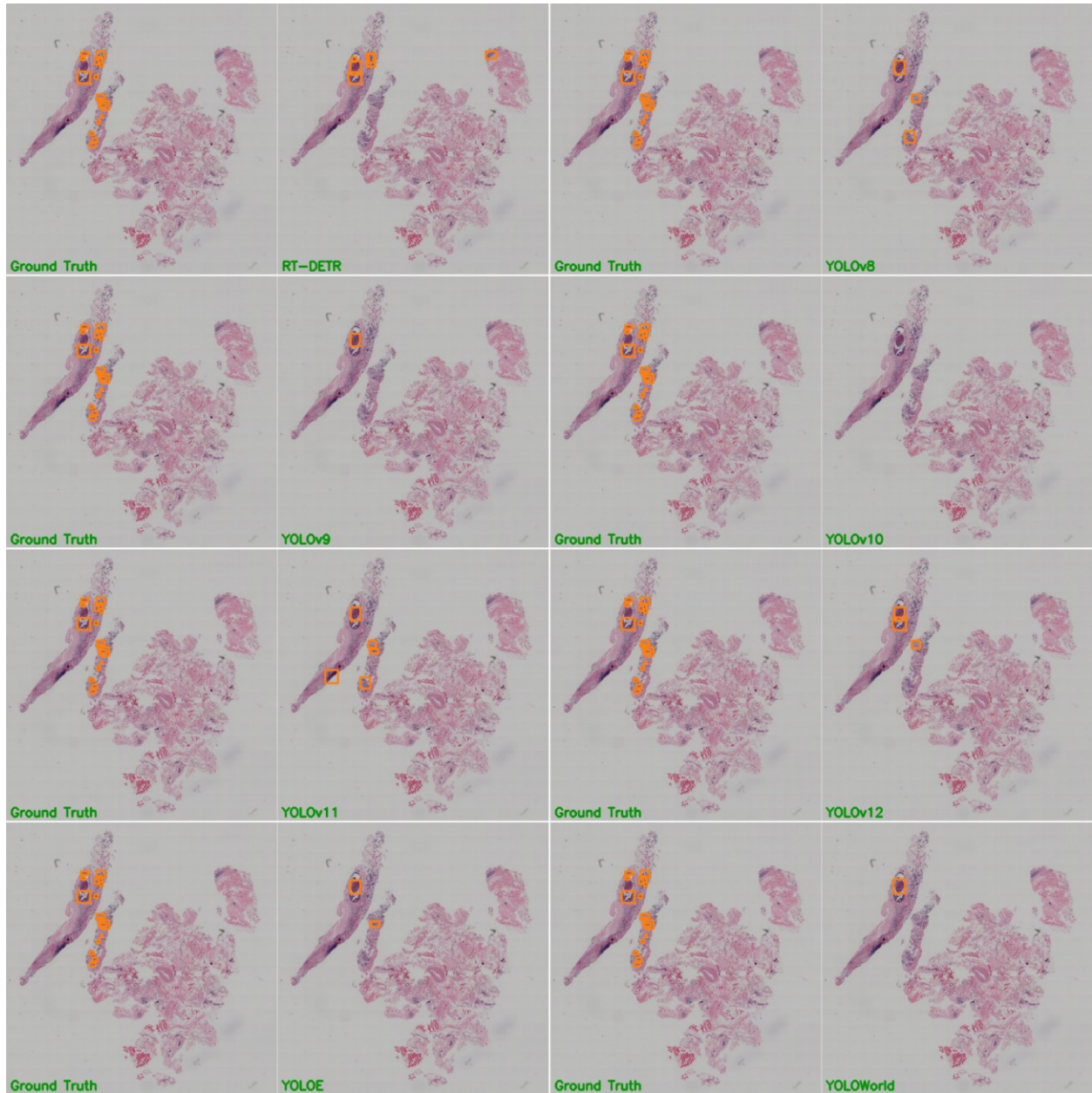


Figure 6 – Example from the BCNB [35] dataset with the true annotations (left) and the predicted bounding boxes (right). The results suggest that the models are unable to comparably recapitulate the bounding boxes observed in the ground truth.

3.2 Nuclei

The Nuclei dataset [10], containing only 143 images, presented unique challenges related to limited training data and small object sizes as shown in Table 3. The results showed more

pronounced differences between models, with some architectures failing completely; nevertheless, the detection accuracy is higher than for the previous dataset.

RT-DETR completely failed on this dataset, achieving zero performance across all metrics. This catastrophic failure suggests that this transformer-based architecture may require more training data or may not be well-suited for the small object sizes typical in histological detection tasks.

YOLOE achieved the highest mAP@50 score of 0.464 and F1 of 0.473, closely followed by YOLOWorld and YOLOv11. The prompt-guided capabilities of YOLOE appeared to provide advantages on this limited dataset, possibly by better leveraging contextual information during training.

Interestingly, YOLOv12 underperformed compared to YOLOv11, achieving only 0.318 mAP@50. However, YOLOv12 showed the highest recall (0.489), suggesting it performed better at detecting all nuclei, but with lower precision. This trade-off indicates that YOLOv12's attention mechanisms may be capable of detecting more potential objects, but with reduced discrimination capability on this small dataset.

Table 3 – Results of object detection models on the Nuclei [10] dataset.

Model/Metric	mAP@50	mAP@50-95	Mean F1	Mean Precision	Mean Recall	Inference Time [ms]	Computational Cost [GFLOPs]
RT-DETR	0	0	0	0	0	1123.8	103.4
YOLOv8	0.362	0.214	0.342	0.504	0.259	301.8	28.4
YOLOv9	0.37	0.226	0.39	0.48	0.329	211.9	26.7
YOLOv10	0.353	0.218	0.379	0.439	0.334	141.3	24.5
YOLOv11	0.402	0.242	0.433	0.47	0.401	155.6	21.3
YOLOv12	0.318	0.193	0.385	0.318	0.489	187.7	21.2
YOLOE	0.464	0.279	0.473	0.566	0.407	212.2	35.3
YOLOv8-Worldv2	0.412	0.257	0.405	0.554	0.319	150.9	32.6

Certain insights can also be gained from the visual inspection of results shown in Figure 7. The lower performance of YOLOv12 may be attributed to its tendency to over-predict resulting in many false positives. In contrast, YOLOv11 and YOLOE produce fewer predictions that align more closely the true bounding boxes.

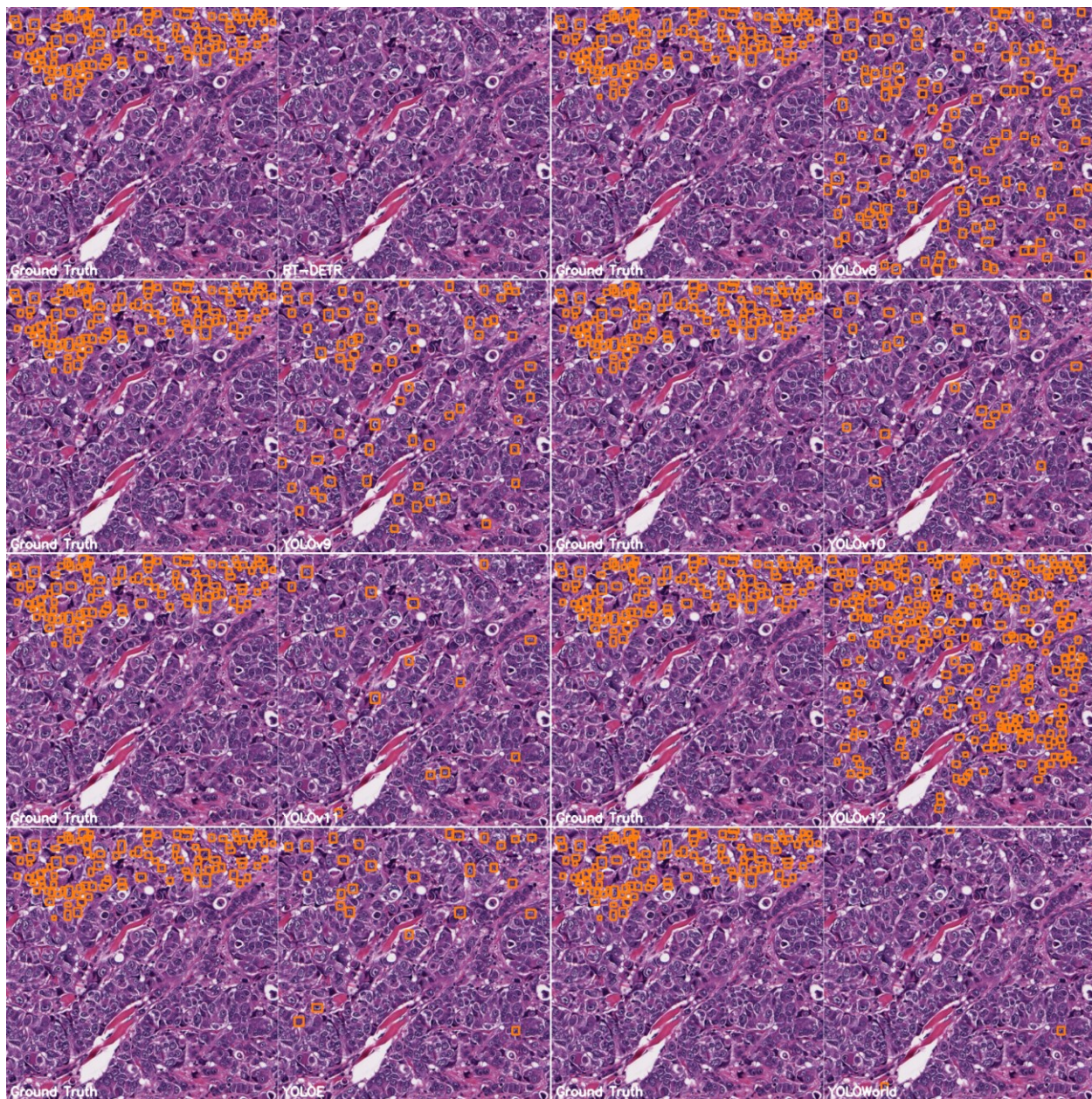


Figure 7 – Example from the Nuclei [10] dataset with the true annotations (left) and the predicted bounding boxes (right). The results suggest that the models are unable to comparably recapitulate the bounding boxes observed in the ground truth (either due to a significant number of false positives, such as YOLO12, or due to a lack of predictions, such as YOLOW).

3.3 TNBC

The TNBC dataset [36] was split across patients into the train, validation and test sets. The 11 patients were split as follows: patients 01 to 09 into the train set, patient 10 into the validation set

and patient 11 into the test set. The chosen splitting of the dataset allowed for an experimental setup where the cross-patient generalization of these models could be evaluated.

The results showed exceptionally high performance across most YOLO models, indicating that triple-negative breast cancer nuclei are relatively well-characterized and distinguishable. Moreover, despite the small number of examples in the dataset, the high performance of models indicates that object size might be a more important factor than the dataset size. The results obtained by each model are presented in Table 4.

Both YOLOv9 and YOLOv12 achieved the highest mAP@50 scores of 0.930, demonstrating excellent detection capability. However, YOLOv12 achieved the highest mAP@50-95 score of 0.676, indicating superior localization accuracy across stricter IoU thresholds. This suggests that YOLOv12's attention mechanisms provide better boundary delineation, which is crucial for accurate quantitative analysis of nuclear morphology. YOLOv12 also achieved the highest F1 score (0.906) and recall (0.890), indicating the best overall balance between precision and recall and a relatively low inference speed of only 189.5 ms, making it a possible choice for real time applications.

RT-DETR showed improved performance on this dataset compared to other datasets, but still significantly underperformed compared to YOLO architectures. The extremely high recall (0.882) but low precision (0.249) suggests RT-DETR was over-detecting, producing many false positives.

Table 4 – Results of object detection models on the TNBC [36] dataset.

Model/Metric	mAP@50	mAP@50-95	Mean F1	Mean Precision	Mean Recall	Inference Time [ms]	Computational Cost [GFLOPs]
RT-DETR	0.334	0.182	0.388	0.249	0.882	1193.6	103.4
YOLOv8	0.924	0.650	0.895	0.921	0.871	315.0	28.4
YOLOv9	0.930	0.597	0.885	0.892	0.878	229.8	26.7
YOLOv10	0.888	0.642	0.861	0.884	0.839	177.7	24.5
YOLOv11	0.923	0.672	0.900	0.936	0.866	167.6	21.3
YOLOv12	0.930	0.676	0.906	0.922	0.890	189.5	21.2
YOLOE	0.920	0.601	0.899	0.921	0.878	241.1	35.3
YOLOv8-Worldv2	0.899	0.649	0.874	0.921	0.831	170.7	32.6

The visual inspection shown in Figure 8 corroborates the insights extracted from the performance metrics. RT-DETR does indeed predict false positives in areas that do not contain pathological features, whilst YOLOv12 has the most accurate bounding boxes. The other YOLO models whilst predicting somewhat accurate bounding boxes tend to predict multiple overlapping boxes for pathological features and miss smaller areas or areas that blend in with the background.

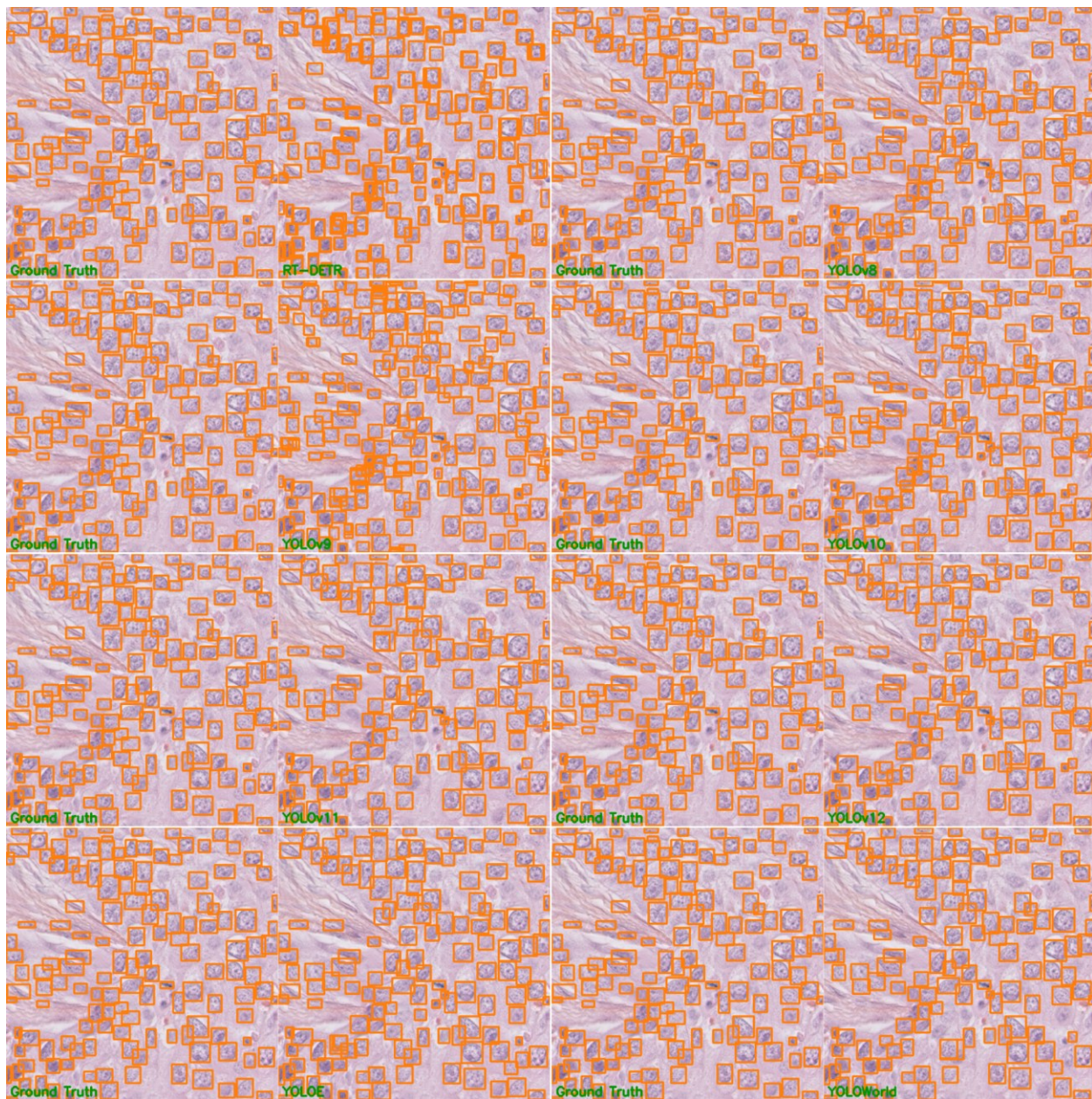


Figure 8 – Example from the TNBC [36] dataset with the true annotations (left) and the predicted bounding boxes (right). The results suggest that all models are able to identify with high accuracy the true bounding boxes with a low number of missed boxes.

3.4 MoNuSAC

The MoNuSAC dataset [37] , with its multi-organ, multi-class detection task, provided the most comprehensive evaluation of model capabilities across diverse tissue types and cellular morphologies. The results across models are shown in Table 5.

YOLOv8-Worldv2 achieved the highest mAP@50 score of 0.780, demonstrating the advantage of open-vocabulary capabilities for multi-class detection tasks. YOLOv11 achieved the second-highest mAP@50 (0.777) and the highest mAP@50-95 (0.510), indicating excellent performance across both detection and localization followed closely by YOLOE. YOLOE demonstrated strong performance with the highest F1 score (0.732) and competitive recall (0.686), suggesting that prompt-guided detection may be particularly beneficial for distinguishing between different cell types.

Surprisingly, YOLOv12 underperformed relative to expectations, achieving only 0.721 mAP@50, lower than YOLOv11, YOLOE, and YOLOv8-Worldv2. However, YOLOv12 showed competitive recall (0.679), suggesting it was detecting most objects but with reduced precision. This pattern suggests that the attention mechanisms in YOLOv12 may require more careful tuning for multi-class detection tasks. RT-DETR again showed poor performance, particularly in mAP@50-95 (0.069), indicating severe localization problems.

Table 5 – Results of object detection models on the MoNuSAC [37] dataset.

Model/Metric	mAP@50	mAP@50-95	Mean F1	Mean Precision	Mean Recall	Inference Time [ms]	Computational Cost [GFLOPs]
RT-DETR	0.306	0.069	0.230	0.449	0.203	1010.4	103.4
YOLOv8	0.748	0.488	0.694	0.781	0.639	190.2	28.4
YOLOv9	0.744	0.463	0.701	0.820	0.626	208.4	26.7
YOLOv10	0.699	0.442	0.649	0.758	0.587	238.8	24.5
YOLOv11	0.777	0.510	0.725	0.806	0.676	211.5	21.3
YOLOv12	0.721	0.477	0.667	0.721	0.679	206.3	21.2
YOLOE	0.773	0.454	0.732	0.804	0.686	253.0	35.3
YOLOv8-Worldv2	0.780	0.483	0.714	0.774	0.693	199.6	32.6

The visual inspection shown in Figure 9 demonstrates that all models have wrong predictions, specifically they produce false positives in areas with similar features to those of true pathological areas. Moreover, several models predicted classes that are not present in the images even overlapping with correct predictions.

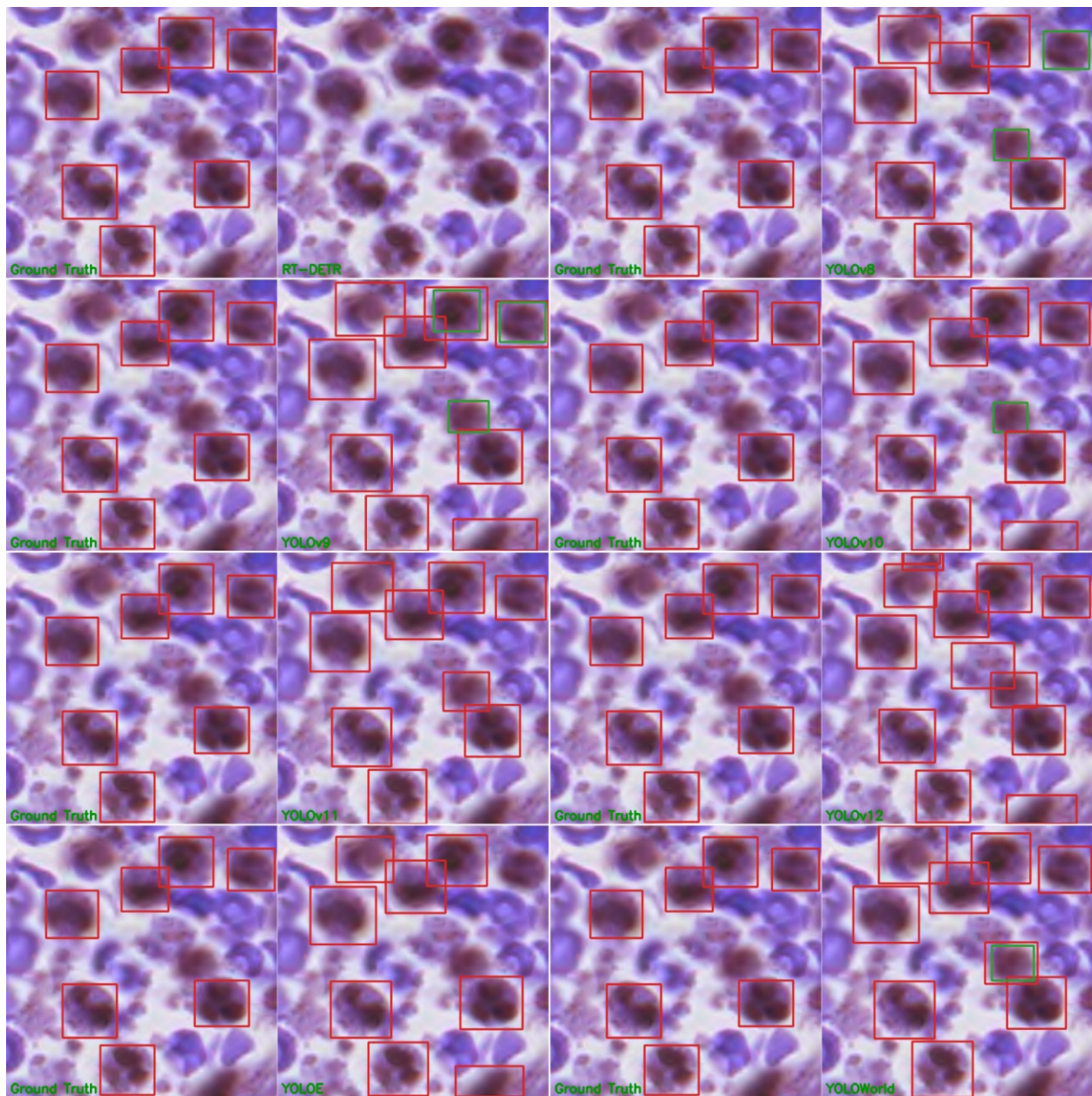


Figure 9 – Example from the MoNuSAC [37] dataset with the true annotations (left) and the predicted bounding boxes (right). The results suggest that all models (except RT-DETR) are able to identify with high accuracy the true bounding boxes; nevertheless, there is a set of annotations that are missed (such as the top-right annotation in YOLOE) or false positives of a different class (such as the green boxes in YOLO8 / YOLO9 / YOLO10 / YOLOW).

3.5 CryoNuSeg

As the CryoNuSeg dataset [38] offers 10 different organs, the dataset was split by organs for training, validation and testing. All organs but the adrenal gland and the larynx have been used for training, whilst the adrenal gland was used for validation and the larynx for testing. The chosen splitting of the dataset allowed for an experimental setup where the cross-organ generalization of these models could be evaluated. The results of this experiment are shown in Table 6.

YOLOv11 achieved the highest mAP@50 score of 0.781 and the highest mAP@50-95 of 0.506, demonstrating superior performance for this cross-organ detection task. YOLOv11 also achieved the best balance between precision (0.767) and recall (0.757), resulting in the highest F1 score (0.762). However, YOLOv12 obtained the same value for mAP@50, while all other metrics were slightly lower. These two models were closely followed by YOLOv8, YOLOE and YOLOWorld on all metrics. YOLOv8 achieved the second-highest mAP@50 (0.773) with the fastest inference time among all models (169.4 ms), demonstrating that earlier YOLO versions can still provide competitive performance on certain tasks. RT-DETR showed catastrophic failure on this dataset with extremely low performance across all metrics (mAP@50 of 0.031), suggesting fundamental incompatibility with the small object sizes.

Table 6 – Results of object detection models on the CryoNuSeg [38] dataset.

Model/Metric	mAP@50	mAP@50-95	Mean F1	Mean Precision	Mean Recall	Inference Time [ms]	Computational Cost [GFLOPs]
RT-DETR	0.031	0.014	0.072	0.056	0.104	778.9	103.4
YOLOv8	0.773	0.469	0.737	0.727	0.748	169.4	28.4
YOLOv9	0.746	0.453	0.711	0.690	0.733	170.2	26.7
YOLOv10	0.675	0.455	0.668	0.694	0.643	186.9	24.5
YOLOv11	0.781	0.506	0.762	0.767	0.757	216.2	21.3
YOLOv12	0.781	0.499	0.744	0.759	0.730	179.0	21.2
YOLOE	0.752	0.423	0.714	0.725	0.703	241.2	35.3
YOLOv8-Worldv2	0.754	0.419	0.724	0.745	0.705	186.9	32.6

Through the data presented in Figure 10, RT-DETR is evidently unable to correctly localize the pathological areas creating much larger boxes than required. Nevertheless, all models tend to over-predict with even smaller areas being identified as pathological although they do not contain such features.

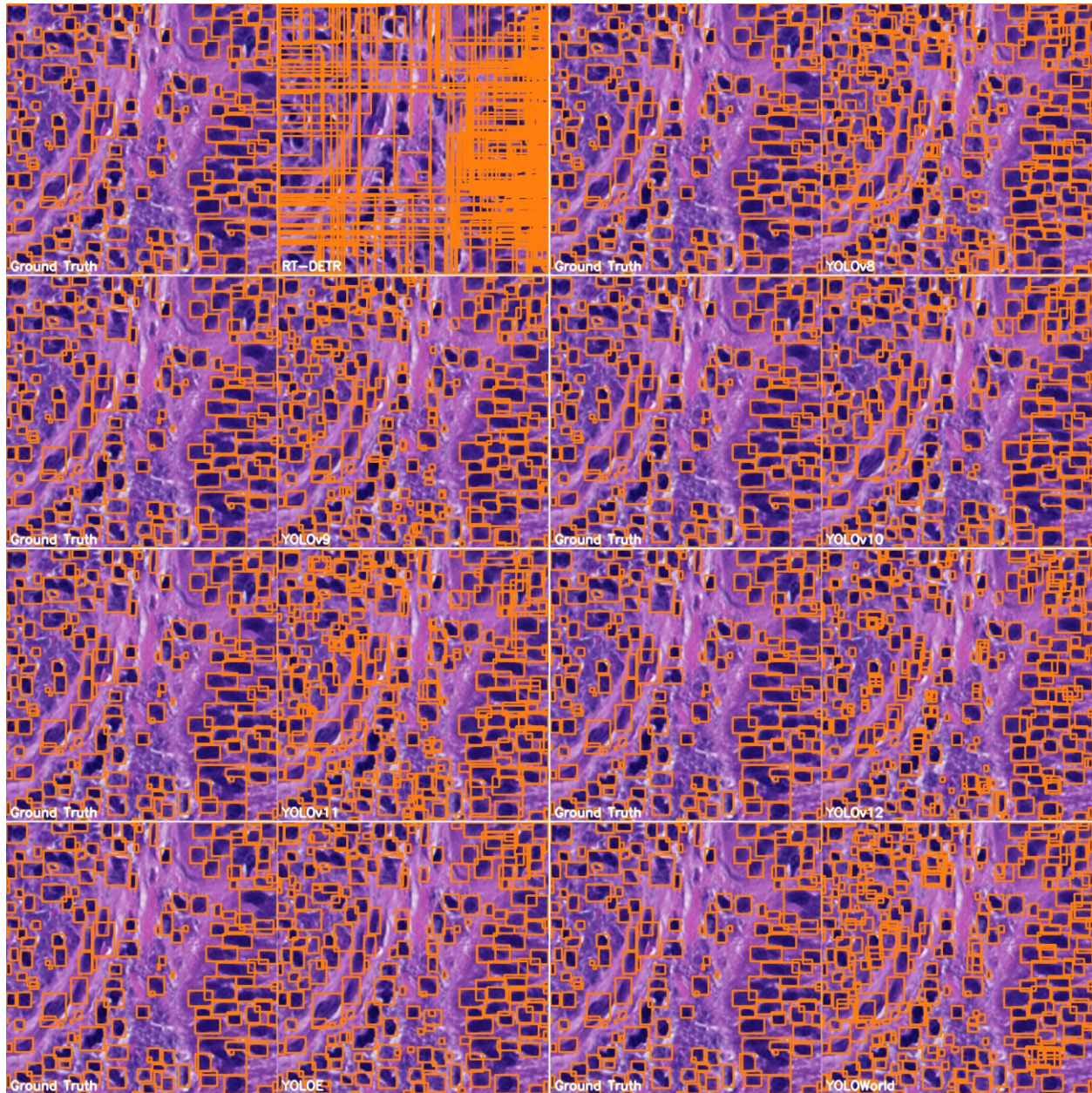


Figure 10 – Example from the CryoNuSeg [38] dataset with the true annotations (left) and the predicted bounding boxes (right). The results suggest that all models (except RT-DETR) are able to identify with high accuracy the true bounding boxes; nevertheless, there are several cases of overlapping boxes which can be attributed to the low value of the IoU-based NMS parameter.

4 Conclusions

The evaluation across five histology datasets with diverse characteristics and limitations revealed several insights regarding the performance characteristics of different object detection

architectures for histopathological applications. The limited training data, the small object sizes and multi-organ multi-class nature are challenges for models even without considering cross-patient and cross-organ generalization.

The high performances of all models on the experimental setup of the TNBC dataset (based on the train-validation-test split) indicate that these object detection models may be capable of cross-patient generalization as the overall performance of the YOLO models have reached a $\text{mAP}@50$ of ~ 0.9 . At the same time, the results obtained through the experimental setup of the CryoNuSeg dataset (based on the train-validation-test split) indicate that these object detection models may even be capable of cross-organ generalization due to the high performance of YOLO models of ~ 0.75 $\text{mAP}@50$. All YOLO models had similar performances on the chosen datasets, yet a few conclusions about their overall ranking can be deduced from the data presented.

YOLOv11 emerged as the most consistent performer across datasets, achieving top or near-top performance on four out of five datasets. Its architecture incorporating C3k2 blocks, SPPF, and C2PSA modules proved effective for histological data. The model showed stability across object sizes and tissue types. YOLOv11 demonstrated through its performance the highest robustness on the CryoNuSeg dataset in a cross-organ generalization setup.

YOLOv12, despite its attention mechanism and state-of-the-art performance on natural datasets, showed variable performance on histological images. While it achieved the highest performance on the TNBC dataset ($\text{mAP}@50-95$: 0.676) and competitive results on BCNB and CryoNuSeg, it underperformed on the Nuclei and MoNuSAC datasets (when compared to other YOLO architectures). This suggests that the attention-centric design may require larger training datasets or specific domain adaptations to fully leverage its capabilities for the small cellular structures typical in histology.

YOLOv10 demonstrated competitive inference times while maintaining reasonable accuracy. YOLOv8 and YOLOv9 showed solid performance across datasets, with YOLOv9 achieving top performance on TNBC ($\text{mAP}@50$: 0.930). These models represent well-optimized architectures that benefit from extensive development and refinement.

YOLOE demonstrated strong performance, particularly on the small Nuclei dataset ($\text{mAP}@50$: 0.464) and multi-class MoNuSAC dataset ($\text{mAP}@50$: 0.773). The prompt-guided detection capabilities appear to provide advantages for scenarios with limited training data or complex classification requirements. YOLOv8-Worldv2 showed particular strength on the multi-class MoNuSAC dataset ($\text{mAP}@50$: 0.780), demonstrating the value of open-vocabulary detection capabilities for applications requiring classification of diverse cell types without extensive retraining.

RT-DETR showed the lowest performance across all datasets, with catastrophic failures on the Nuclei and CryoNuSeg datasets ($\text{mAP}@50$: 0 and 0.031, respectively). The transformer-based architecture appears fundamentally unsuitable for the small object sizes and limited training data of histological images. The extremely high inference times (778.9-1193.6 ms) further limit its clinical applicability. YOLO architectures (126.7-315.0 ms) demonstrated 3-10 times faster inference compared to RT-DETR, while achieving significantly higher detection accuracy.

Overall, based on the similar performances of models across datasets (except RT-DETR), an empirical observation is that object sizes are a better predictor of performance than dataset size. This can be inferred from the significantly lower performance of all models on the BCNB (1058 images) and Nuclei (143 images) datasets, while at the same time the models had a high performance on the other datasets (30, 50 and 209 images) which had considerably lower dataset sizes.

Future research should focus on several areas to advance the automated analysis of histopathological images. First, the investigation of learning approaches to address rare pathologies where only limited data can be obtained, such as few-shot and zero-shot learning. Our results with YOLOE and YOLO-World suggest that prompt-guided and open-vocabulary approaches show promise for these scenarios. Second, the development of explainable AI techniques specifically designed for histopathological applications is essential for clinical acceptance and approval. Visualization of feature importance, decision boundaries, and inner mechanisms of models would help pathologists understand and trust the automated systems. Our analyses provide a starting point, but clinical deployment requires extensive validation beyond standardized datasets.

In conclusion, while recent advances in object detection architectures have provided powerful tools for histopathological image analysis, careful consideration of specific application requirements, dataset characteristics, and practical deployment constraints are required for clinical implementation. These findings provide guidance for researchers and clinicians for the development of automated histopathological analysis systems and establish performance benchmarks for future innovations in this area.

Data Availability

The datasets used in this work are openly available and can be found at:

- BCNB [35]: <https://bcnb.grand-challenge.org/>
- Nuclei [10]: <https://andrewjanowczyk.com/use-case-1-nuclei-segmentation/>
- TNBC [36]: https://zenodo.org/records/2579118#.Yt5FWt_RaUk
- MoNuSAC [37]: <https://monusac-2020.grand-challenge.org/>
- CryoNuSeg [38]: <https://www.kaggle.com/datasets/ipateam/segmentation-of-nuclei-in-cryosectioned-he-images>

Code Availability

The code that supports the findings of this work was written in Python and is openly available at: <https://github.com/ArdeleanRichard/YOLO-Histology/>.

Competing Interests

The authors have declared that no competing interests exist.

Authors' contributions

Conceptualization, E.-R.A. and D.L.; methodology, E.-R.A. and D.L.; software, E.-R.A. and D.L.; validation, E.-R.A. and D.L.; formal analysis, E.-R.A. and D.L.; investigation, E.-R.A. and D.L.; data curation, E.-R.A. and D.L.; writing—original draft preparation, E.-R.A. and D.L.; writing—review and editing, E.-R.A. and D.L.; visualization, E.-R.A. and D.L.; supervision, E.-R.A. and D.L.; administration, E.-R.A. and D.L. All authors have read and agreed to the published version of the manuscript.

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